Phase 5: Enable Tool Usage

1. Objective
This document provides the complete implementation for Phase 5.
Previous phases introduced: • Rule-based banking support • LLM reasoning • Safety-aware prompts • Retrieval-Augmented Generation (RAG). Phase 5 upgrades the banking AI agent further by enabling controlled tool usage. The agent can now intelligently select and use tools while still respecting strict banking safety requirements.

2. Architecture:

```
Customer Query
      ↓
Intent Detection
      ↓
Tool Needed?
 ┌───────────────┐
 │ Yes           │ No
 ▼               ▼
Call Tool      RAG Search
 ▼               ▼
Tool Result   Retrieved Docs
      \         /
       \       /
        ▼     ▼
          LLM
           ↓
        Response
```

3. Defined two tools

For banking support, a good choice is:

Tool 1: EMI Calculator

Calculates monthly loan payments.

Tool 2: Loan Eligibility Estimator

Simple eligibility check.

4. Demonstrating Correct Tool Usage
The following demonstrates the correct usage of the tool

```
Customer: What is the EMI for a ₹5 lakh loan at 8.5% for 5 years?

Agent:
[TOOL USED: emi]
Estimated EMI: ₹10258.27

Customer:
```

5. Demonstrating Failed Tool Call
The query did not require a calculation tool. The failure occurred because the routing logic incorrectly matched the request to the EMI calculator. This demonstrates a common tool-selection error and highlights the need for more robust intent classification.

```
labsuser@jupyter:~$ python phase_5_enable_tool_usage.py
/usr/local/lib/python3.10/site-packages/trio/_core/_multierror.py:511: Runt
 but this means MultiErrors will not show full tracebacks.
  warnings.warn(
Loaded 8 banking documents
======================================================================
Banking AI Support & Advisory Agent
Phase 5 – Tool Enabled Agent
======================================================================


============================================================
FAILED TOOL CALL DEMONSTRATION
============================================================

Query: What documents are required for a home loan?

Incorrect Tool Selected:
EMI Calculator
Result: Estimated EMI: ₹10258.27

Explanation:
The query required policy retrieval but the EMI tool was selected.
```

6.Safeguards
Have added the following safeguards
(1) The system must refuse money transfers.
(2) Tool Whitelist
(3) Missing Parameters
(4) Prevent repeated tool calls.

Following are some of the screenshot of the safeguards for some of the customer queries and one edge case in the end.

```
Customer: What is the wealth management fee?

Agent:
The wealth management fee is not provided in the available context.

Customer: Transfer ₹20,000 to account 123456789

Agent:
I cannot perform banking transactions or money transfers.

Customer: What is my account balance?

Agent:
I’m sorry, but I don’t have access to your account balance from the provided context. Please check your banking app, online banking, ATM, or contact your bank’s customer support for your current bal
ance.

Customer: Tell me about loans.

Agent:
We currently have information available for **Home Loans**.

**Home Loan Interest Rate:**
- **8.25% per annum**

Please let me know what specific loan details you’d like to know, such as eligibility, documents, EMI, tenure, or application process.
```

7. Limitations

1. Rule-Based Tool Selection

Tool routing is currently implemented using simple keyword matching. Complex or ambiguous user queries may result in incorrect tool selection. Production systems typically use intent classification models or LLM-based function calling to improve routing accuracy.

2. Limited Tool Coverage

The agent currently supports only EMI calculation and loan eligibility estimation. Many real banking scenarios require additional tools such as transaction lookup, fraud reporting workflows, policy databases, account verification systems, and human support escalation mechanisms.

Conclusion

Phase 5 extended the Banking AI Support & Advisory Agent by introducing tool usage capabilities. Two banking-related tools, an EMI calculator and a loan eligibility estimator, were integrated into the agent workflow. The system was enhanced with routing logic to select appropriate tools based on user intent and demonstrated both successful and failed tool invocations. Safeguards including tool restrictions, parameter validation, and loop prevention were implemented to improve safety and reliability. This phase moves the agent beyond information retrieval by enabling it to perform specific banking-related actions while maintaining controlled and explainable behaviour.